Holo-Lab: An Offline-First Augmented Reality Framework for Resilient STEM Education in Low-Resource Schools

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***Abstract*—Rural STEM education can be disrupted by limited laboratory infrastructure, unreliable Internet connectivity, and unequal access to personal digital devices. Holo-Lab proposes an offline-first augmented reality (AR) learning framework intended to improve the resilience and continuity of experiential science education in such environments. The system transforms diagrams in existing NCERT science textbooks into interactive AR experiences and supports teacher-led or small-group learning through shared devices. Educational content is prepared as an offline installation bundle, distributed locally through mechanisms such as school Wi-Fi, teacher devices, USB/SD transfer, or community learning centres, and used without continuous connectivity after installation. Lightweight local learning analytics support concept-level feedback, while periodic synchronization can be used when connectivity becomes available. The framework also includes multilingual guidance and a proposed Teach Me active-recall activity. A functional frontend prototype has been developed, but technical, usability, and learning-performance measurements are future validation tasks. The work positions offline AR as a resilient educational support layer rather than a replacement for teachers or physical laboratories.**

***Index Terms*—resilient education, offline-first systems, augmented reality, rural education, sustainable technology, STEM learning, shared-device computing, digital inclusion**

I. Introduction

Access to schooling has expanded considerably across India; however, school attendance does not necessarily guarantee conceptual or practical learning. Rural and low-income government schools may face limitations in laboratory infrastructure, equipment, recurring consumables, and access to technology-supported learning environments. Conventional digital learning platforms also often assume reliable Internet connectivity and individual access to smartphones or computers. These assumptions are difficult to satisfy uniformly in low-resource communities.

UNICEF has highlighted digital-access inequalities affecting children in remote and impoverished communities, while later guidance continues to identify infrastructure and access inequalities in education [8], [7]. The Annual Status of Education Report (ASER) provides large-scale evidence concerning learning conditions in rural India and has increasingly examined digital skills and smartphone usage among adolescents [6].

These conditions motivate educational technologies that do not simply digitize textbooks but improve how students interact with concepts. Augmented reality (AR) is relevant because it can overlay interactive digital representations onto physical learning materials. Previous research reports educational potential for AR while also identifying usability and technical challenges [1], [2]. For STEM learning, interactive three-dimensional representations can make processes difficult to understand from two-dimensional diagrams more observable and manipulable [3].

Holo-Lab therefore proposes an offline-first AR learning framework in which existing NCERT science textbooks act as physical anchors for interactive digital learning experiences. The objective is not to replace teachers, textbooks, or physical laboratories, but to provide an additional experiential learning layer for schools where laboratory resources or digital infrastructure are limited.

II. Conference-Oriented Contributions

The main contributions are: (1) an offline-first AR architecture for resilient STEM learning; (2) a local content-distribution and periodic-synchronization model; (3) a shared-device classroom strategy that reduces dependence on one-to-one hardware; (4) textbook-anchored AR for experiential learning; (5) local concept-level analytics and Teach Me reinforcement; and (6) a privacy-aware deployment model for low-resource schools.

III. Resilience-Oriented Design

Educational resilience requires a system to continue providing useful learning functions when normal infrastructure assumptions fail. Holo-Lab treats connectivity loss, limited devices, limited laboratory access, language diversity, and teacher workload as operational constraints.

TABLE I
RESILIENCE REQUIREMENTS AND HOLO-LAB RESPONSES

Constraint	Holo-Lab response
Unreliable Internet	Offline-first learning after installation
Limited devices	Shared-device teacher and small-group modes
Limited laboratories	AR-based supplementary visualization
High download cost	Local installation and distribution
Language diversity	Multilingual text/audio roadmap
Teacher workload	Concept-level learning-gap indicators
Data concerns	Data minimization and pseudonymous analytics

A. Operational Continuity

The architecture separates initial content distribution, local learning operation, and later synchronization. This prevents loss of core learning functionality when the network is unavailable.

B. Sustainable Infrastructure Use

The framework reuses existing textbooks and shared devices instead of requiring new textbooks, dedicated laboratories, or one-device-per-student deployment. The cost discussion therefore focuses on reducing additional infrastructure requirements, not on claiming zero deployment cost.

IV. System Description

A. Holo-Lab Concept

Holo-Lab converts selected diagrams in existing NCERT science textbooks into interactive AR learning experiences. The intended sequence is

$$\begin{aligned}\text{Textbook} &\rightarrow \text{Camera Scan} \rightarrow \text{Target Recognition} \\ &\rightarrow \text{3D Model} \rightarrow \text{Interactive Learning} \rightarrow \text{Quiz}\end{aligned} \tag{1}$$

For example, a textbook representation of the human eye can be augmented with an interactive three-dimensional representation showing the lens, retina, and path of light.

B. Solo Mode

Solo Mode supports individual or small-group learning. A learner can scan a supported textbook diagram, view the AR model, rotate and explore it, access voice-guided explanations, complete short concept quizzes, and receive additional learning prompts based on quiz performance. The design supports shared or family-owned devices rather than assuming individual smartphone ownership.

C. Classroom Mode

Classroom Mode addresses the limitation of one-device-per-student deployment. A teacher can use a smartphone or tablet to display the AR activity on a classroom display or projector where available. Without projection infrastructure, the same content can be used directly on the teacher's device in small-group settings.

V. Technical Architecture

The proposed system consists of six layers.

A. Physical Learning Layer

The NCERT textbook acts as the physical learning anchor. Selected textbook diagrams provide visual targets for the AR experience.

B. Recognition Layer

The camera captures the textbook page and identifies a supported visual target.

C. AR Rendering Layer

After recognition, the corresponding optimized three-dimensional educational asset is loaded for interactive visualization.

D. Learning Interaction Layer

Students interact with three-dimensional models, animations, explanations, experiment variables, and quizzes.

E. Local Intelligence Layer

Quiz results can be converted into concept-level indicators:

$$\begin{aligned}\text{Quiz} &\rightarrow \text{Concept Score} \\ &\rightarrow \text{Weak Concept Detection} \rightarrow \text{Scaffolding Activity}\end{aligned} \tag{2}$$

F. Teacher Analytics Layer

Aggregated concept-level information can be made available to teachers through local synchronization or periodic synchronization when connectivity becomes available.

VI. Offline-First Architecture

A central design principle is that continuous Internet connectivity should not be necessary for learning after content installation. An offline-first application nevertheless requires initial content distribution.

A. Content Preparation

The application and educational assets are packaged into an offline installation bundle.

B. Local Distribution

The package can be distributed through school Wi-Fi, teacher devices, USB or SD-card transfer, and local community or digital learning centres.

C. Offline Operation

After installation, supported AR content, quizzes, and local learning functions are designed to operate without continuous Internet connectivity.

TABLE II
TARGET HARDWARE AND PERFORMANCE SPECIFICATIONS

Metric	Design Target
Target device RAM	Approximately 2 GB
AR loading time	< 2.0 s
Target frame rate	≥ 30 FPS
Target application size	< 80 MB
Target crash rate	$< 1.0\%$

TABLE III
PROPOSED CURRICULUM CONTENT ROADMAP

Class Group	Experiments	3D Models	Languages
Classes 6–7	18	45	8
Classes 8–9	24	63	8
Class 10	20	52	8
Total	62	160	8

D. Periodic Synchronization

When connectivity becomes available, lightweight application updates and aggregated learning reports may be synchronized rather than requiring the complete learning package to be downloaded again.

VII. Hardware and Asset Optimization

The initial design targets entry-level Android devices with approximately 2 GB of RAM. This is a design target rather than a measured minimum. No formal hardware benchmark was conducted for the present study, so measured performance values are not reported.

Future evaluation will benchmark startup time, AR loading time, frame rate, memory consumption, application size, battery consumption, and crash rate on representative entry-level Android devices.

VIII. Curriculum Content Model

The proposed roadmap targets 62 experiments and 160 three-dimensional models across Classes 6–10. These are proposed roadmap values, not claims that all assets are already developed. The target language roadmap includes Tamil, Hindi, Telugu, Kannada, Bengali, Marathi, Gujarati, and English.

IX. Teach Me Active-Recall Mode

The "Teach Me" mode is designed around active explanation. The student explains a scientific concept in their own words. The proposed pipeline is

Student Speech $\rightarrow$ Speech Recognition $\rightarrow$ Concept Identification $\rightarrow$Coverage Score $\rightarrow$ Feedback

For example, for refraction, the system may evaluate whether important concepts such as incident ray, refracted ray, angle of incidence, and angle of refraction are represented.

The current prototype defines the concept-level workflow, while fully multilingual offline speech recognition remains part of the extended implementation and requires dedicated validation.

X. Multilingual and Accessible Design

Holo-Lab uses text instructions, audio guidance, visual cues, simplified navigation, and reduced dependence on long textual explanations. The eight-language roadmap represents proposed localization scope rather than a claim that all language functions have been fully implemented and validated.

XI. Deployment Model

A typical deployment can operate through a teacher's smartphone or tablet, shared classroom displays where available, small-group learning using one shared device, and local content-distribution hubs. The system is designed around shared devices rather than a one-device-per-student assumption.

XII. Prototype and Application Interface

The current Holo-Lab development version includes the main user-interface flow for authentication, dashboard access, experiment browsing, AR scanning, assessment, profile management, and learning-progress views. In the same progressive presentation style used in Paper 32, the application screens are distributed across six figures so that the reviewer can follow the user workflow from entry to learning activity and progress monitoring. The screenshots document the current development state; they are not presented as evidence of experimentally validated learning gains or measured system performance. The complete application, including full AR content, offline deployment, multilingual interaction, and learning analytics, remains under development.

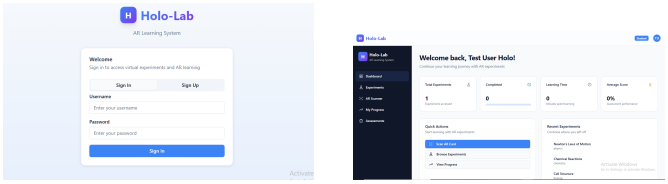


Fig. 1. Holo-Lab authentication interface (left) and main dashboard (right).

The authentication screen provides entry to the application, while the dashboard provides the main navigation point for learning activities and experiment access.

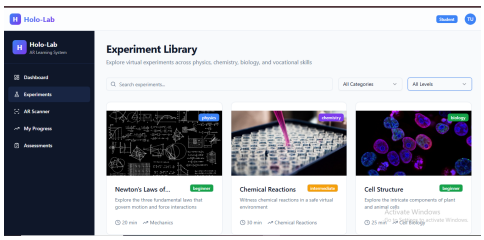


Fig. 2. Holo-Lab experiment and STEM learning interface.

The experiment interface presents the available learning activity and provides the starting point for the intended textbook-linked interactive learning workflow.

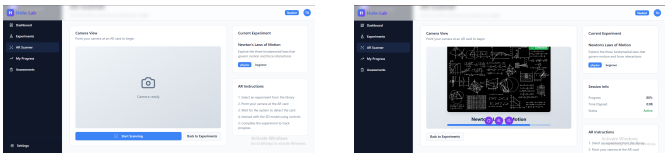


Fig. 3. Textbook/camera scanning interface (left) and recognized target interface (right).

The camera-based workflow is intended to recognize a supported textbook target before loading the corresponding AR learning experience. The detected-target screen represents the transition from recognition to interactive content.

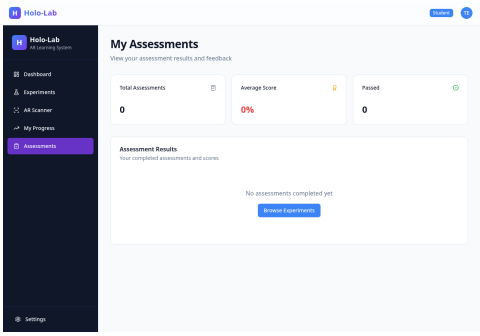


Fig. 4. Holo-Lab assessment interface.

The assessment interface represents the quiz and concept-checking stage of the learning workflow. It provides the interface basis for the proposed adaptive assessment and local concept-level feedback.

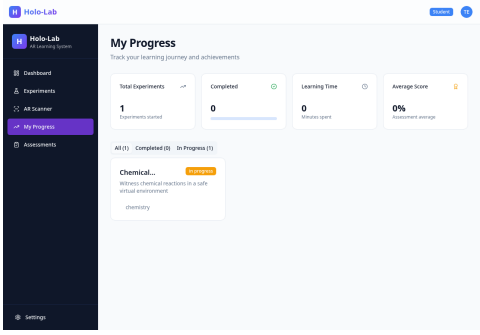


Fig. 5. Holo-Lab learning-progress interface.

The progress interface provides a user-facing view of learning activity and forms part of the proposed learning-analytics workflow.

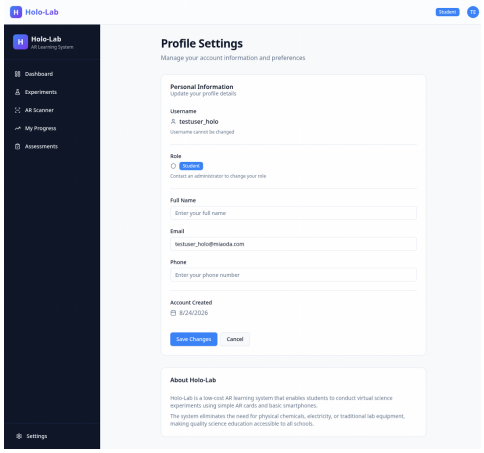


Fig. 6. Holo-Lab user profile interface.

The profile interface provides user-oriented controls and complements the privacy and settings design described later in the paper.

XIII. Data Privacy and Child Safety

Guest Mode is intended for shared-device use and minimizes persistent student identity requirements. Where persistent analytics are enabled, the design uses pseudonymous student identifiers and restricts learning reports to authorized users. Data-minimization principles are particularly important for children. Raw speech recordings should not be retained by default unless required for an explicitly defined educational purpose and appropriate consent procedures are established. Privacy controls and data-management practices must be validated with schools before large-scale deployment.

XIV. Teacher Learning-Gap Dashboard

Holo-Lab proposes converting quiz performance into concept-level indicators. Instead of reporting only that a group performed poorly, the system can identify topics such as refraction or electric circuits that require additional instructional support:

$$\text{Student Responses} \rightarrow \text{Concept Mapping} \rightarrow \text{Learning Gap} \rightarrow \textit{Teacher Feedback} \tag{3}$$

XV. Evaluation Plan

No formal quantitative user study or hardware benchmark was conducted for the present version; therefore, no fabricated performance or learning-outcome values are reported.

A. Technical Evaluation

Future testing will measure AR loading time, frame rate, peak memory usage, application size, battery consumption, and crash frequency on representative entry-level Android devices.

B. Learning Evaluation

A future pilot can compare pre-intervention and post-intervention concept scores:

$$G = S_{\mathrm{post}} - S_{\mathrm{pre}},$$

where S_{pre} and S_{post} are pre-test and post-test scores.

C. Usability Evaluation

Usability can be evaluated with the System Usability Scale (SUS), task completion rate, and qualitative student/teacher feedback. No numerical usability or learning results are reported because such measurements have not yet been conducted.

XVI. Conclusion

Holo-Lab contributes to resilient education by separating learning continuity from continuous connectivity. Its combination of offline content, local interaction, shared devices, local analytics, and periodic synchronization is intended to provide a practical support layer for STEM education in infrastructure-constrained schools.

References

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